

376.082 MANIPULATION AND LOCOMOTION – HW2

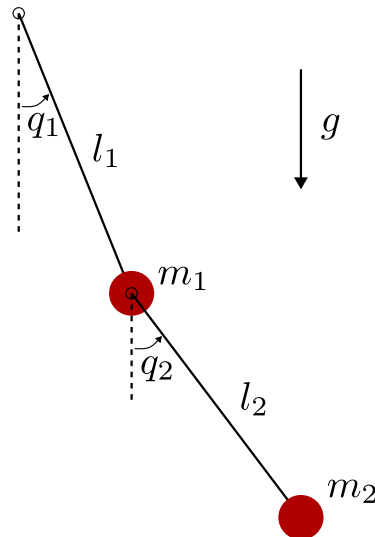
ACIN/RSL, TU WIEN

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Notes:

- Please submit your homework in a concise and reader-friendly form. Avoid unnecessary outputs in your report, navigate the reader, and present the results cleanly. Unorganized or unreadable submissions are subject to point deduction.
- Your report should be self-contained, completely separated from your source code, and submitted in PDF format. All your figures must be readable and included in your report. Even though we keep the right to examine your source code, this is only done if further justification is necessary for your submission.
- The language of the submission is English.
- Collaboration is encouraged. On the other hand, each submission should be unique. The exact or very similar submissions are subject to point deduction from all parties regardless of who helped whom.
- For all questions regarding the homework and projects contact “ml@acin.tuwien.ac.at”.

Question 1: Adaptive Control



Consider double pendulum dynamics

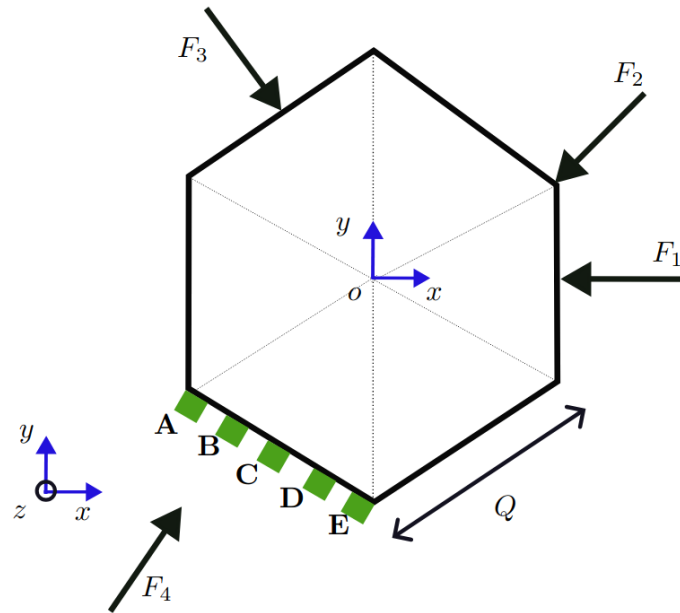
$$\begin{bmatrix} (m_1 + m_2)l_1^2 & m_2l_1l_2\cos(q_2 - q_1) \\ m_2l_1l_2\cos(q_2 - q_1) & m_2l_2^2 \end{bmatrix} \begin{bmatrix} \ddot{q}_1 \\ \ddot{q}_2 \end{bmatrix} + \begin{bmatrix} 0 & -m_2l_1l_2\sin(q_2 - q_1)\dot{q}_2 \\ m_2l_1l_2\sin(q_2 - q_1)\dot{q}_1 & 0 \end{bmatrix} \begin{bmatrix} \dot{q}_1 \\ \dot{q}_2 \end{bmatrix} + \begin{bmatrix} (m_1 + m_2)gl_1\sin(q_1) \\ m_2gl_2\sin(q_2) \end{bmatrix} = \begin{bmatrix} \tau_1 \\ \tau_2 \end{bmatrix},$$

where $g = 9.81m/s^2$ and $\{m, l\}$ are system parameters.

1. Let $m_{1,2} = 1 \text{ kg}$ and $l_{1,2} = 1 \text{ m}$. Formulate a trajectory tracking controller for $q_1(t) = \frac{\pi}{8} - \frac{\pi}{8} \cos(t)$ and $q_2 = \dot{q}_2(t) = \ddot{q}_2(t) = 0$. Simulate the system's response for initial condition $\mathbf{q}(0) = [0, 0]$, $\dot{\mathbf{q}}(0) = [0, 0]$. Justify if there is a tracking error.
2. Update the system parameters as $m_1 = 4 \text{ kg}$ and $m_2 = 2 \text{ kg}$ in the system dynamics such that there is a modeling error in masses. Perform the following:
 - (a) Choose a proper minimal parameter set.
 - (b) Write the regressor matrix.
 - (c) Implement an adaptive controller with the given trajectories. Justify the trajectory and parameter convergence separately. Comment why if not converged. (Choose $K_s = \text{diag}(10, 10)$, $K = \text{diag}(10, 10)$, and $\Gamma = \text{diag}(10, 10)$)
3. Keep the control parameters constant and modify the reference trajectory given in the first step such that the minimal set of parameters converges to their actual values. Explain why they converge with this new trajectory.

Question 2: Grasping

Assume a 4-finger planar grasp with point contact without friction for the following object:



1. Contact points for fingers 1, 2, and 3 corresponding to the forces F_1 , F_2 , and F_3 , respectively, are defined as shown in the figure. Choose an appropriate contact point for finger 4 from the set A,B,C,D,E to achieve force closure. All these five points are equidistant from each other. Note that all the forces F_1 , F_2 , F_3 , and F_4 are applied perpendicular to the edge in contact.
2. Compute the grasp matrix G for your choice.

3. Compute feasible contact force $F_i \geq 0$ to realize the following resultant object forces, where $F_0 = \begin{bmatrix} F_x & F_y & \tau \end{bmatrix}^T$ (with $Q = 2$)

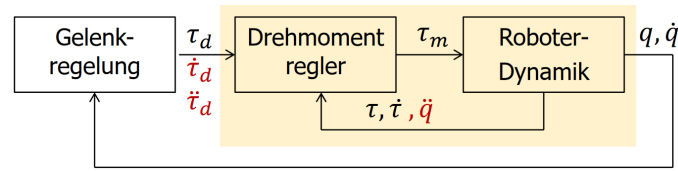
$$F_0 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, F_0 = \begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix}, F_0 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, F_0 = \begin{bmatrix} 0 \\ -1 \\ 0 \end{bmatrix}, F_0 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, F_0 = \begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix} \quad (1)$$

Question 3: Elastic Joint

Consider a single elastic joint system without gravity, described by the equations of motion:

$$\begin{aligned} m\ddot{q} &= k(\theta - q) \\ b\ddot{\theta} + k(\theta - q) &= \tau_m, \end{aligned} \quad (2)$$

where q represents the link position, θ represents the motor position, and τ_m is the applied motor torque. The system parameters are link inertia $m = 2$, motor inertia $b = 0.75$ and spring stiffness $k = 300$. Design a cascaded control system with an outer PD+ and an inner joint torque controller.



1. Select suitable controller gains and simulate the system response to the step input

$$q_d(t) = \begin{cases} 0, & t < t_d, \\ 0.5, & t \geq t_d, \end{cases} \quad (3)$$

so that the closed-loop system exhibits an approximately critically damped response.

2. Repeat the simulation for the following parameter variations:

- reduced stiffness: $k = 10$,
- increased motor inertia: $b = 15$.

Plot the following simulation results for each variation:

- The actual responses $q(t)$ and $\theta(t)$ compared to q_d .
- The control input $\tau_m(t)$.

Discuss how these changes affect the tracking performance, the required motor torque, and the difference between the joint position $q(t)$ and the motor position $\theta(t)$ during the transient response.

3. Extend the model by adding viscous friction, once at the joint and once at the motor. Repeat the simulation for both cases with the initial parameters and compare the results. Plot $q(t)$, $\theta(t)$, and $\tau_m(t)$, and briefly discuss the influence of the friction location on the transient response, the tracking performance, and the required motor torque.